CURRICULUM VITAE
WILLIAM CHAN

william.chan@tuwien.ac.at
Institute for Discrete Mathematics and Geometry
Technische Universität Wien
Wiedner Hauptstraße 8-10/104
1040 Wien, Austria

1. Education

- Habilitation, Mathematics, Technische Universität Wien, 2026. Thesis: "Infinite Cardinal Exponentiation".
- Ph.D, Mathematics, California Institute of Technology, 2017. Advisor: Alexander Kechris. Thesis: "Aspects of Definability for Equivalence Relations".
- B.A., Mathematics, University of Chicago, 2012.

2. Employment

- Technische Universität Wien, Postdoctoral Fellow; 2023- .
- University of North Texas, Postdoctoral Research Associate, 2022-2023.
- Carnegie Mellon University, Fang Zheng Postdoctoral Fellow, 2020-2022.
- University of North Texas, NSF Postdoctoral Fellow and Research Assistant Professor, 2017-2020.

3. Grants

- FWF 10.55776/PAT1500026. Principal Investigator Project. "Size of Infinite Mathematical Objects." Technische Universität Wien. Approved and pending start date.
- FWF 10.55776/PAT1651926. Principal Investigator Project. "Mathematical Structures in the Regular Universe." Technische Universität Wien. 2026-2030.
- NSF DMS-1703708. NSF Mathematical Sciences Postdoctoral Research Fellowship. "Descriptive Set Theory and Definable Sets". The University of North Texas. 2017-2021.

4. Papers

(1) "Cardinality of the Set of Bounded Subsets of a Cardinal". In preparation.
(2) "The ABCD Hypothesis". *Proceedings of the National Academy of Sciences of the United States of America*. 123 (44), e2621223123 (2026).
(3) "The Size of a Set from the Size of Its Countable Power Set". *Pacific Journal of Mathematics*, 344 (2026), no 1, 219-239.
(4) "Coding and Anticoding of a Cardinal by Bounded Subsets of the Cardinal". *Journal of the London Mathematical Society*, 113 (2026): e70501.
(5) "Boldface GCH Below the First Uncountable Limit Cardinal". With Stephen Jackson and Nam Trang. *Annals of Pure and Applied Logic*, 177 (2026), no 10.
(6) "Size of Pieces in Decompositions into the First Uncountable Cardinal Many Pieces". With Stephen Jackson and Nam Trang. To appear in *The Journal of Symbolic Logic*.
(7) "∞-Borel Codes in Natural Models of AD^+". With Stephen Jackson and Nam Trang. Submitted.
(8) "Almost Disjoint Families under Determinacy". With Stephen Jackson and Nam Trang. *Advances in Mathematics*, 347 (2024).
(9) "Definable Combinatorics of Strong Partition Cardinals". In preparation.
(10) "Almost Everywhere Behavior of Functions According to Partition Measures". With Stephen Jackson and Nam Trang. *Forum of Mathematics, Sigma*, 12 (2024): e16, 1-36.

(11) "Countable Length Everywhere Club Uniformization". With Stephen Jackson and Nam Trang. *The Journal of Symbolic Logic*, 88 (2023), no. 4, 1556-1572.
(12) "The Size of the Class of Countable Sequences of Ordinals". With Stephen Jackson and Nam Trang. *Transactions of the American Mathematical Society*, 375 3 (2022), 1725-1743.
(13) "A Lipschitz Determinacy Principle Equivalent to Weak König Lemma". *Annals of Pure and Applied Logic*, 174 (2023), no 3.
(14) "Equivalence of Codes for Countable Sets of Reals". *Canadian Mathematical Bulletin*, 64, no. 3 (2021), 580-590.
(15) "More Definable Combinatorics around the First and Second Uncountable Cardinal". With Stephen Jackson and Nam Trang. *Journal of Mathematical Logic*, 23 (2023), No. 3.
(16) "Definable Combinatorics at the First Uncountable Cardinal". With Stephen Jackson. *Transactions of the American Mathematical Society*, 374 3 (2021), 2035-2056.
(17) "Applications of Infinity-Borel Codes to Definability and Definable Cardinals". With Stephen Jackson. *Fundamenta Mathematicae*, 265 (2024), 215-258.
(18) "Bounds on Continuous Scott Rank". With Ruiyuan Chen. *Proceedings of the American Mathematical Society*, 148 8 (2020), 3591-3605.
(19) "Bounds on Scott Ranks of Some Polish Metric Spaces". *Journal of Mathematical Logic*, 21 (2021), no 1.
(20) "An Introduction to Combinatorics of Determinacy". *Trends in Set Theory*, *Contemporary Mathematics*, 752 (2020), 21-75.
(21) "The Destruction of the Axiom of Determinacy by Forcing on $\mathbb{R}$ when Θ is Regular". With Stephen Jackson. *Israel Journal of Mathematics*, 241 (2021), 119-138.
(22) "Cardinality of Wellordered Disjoint Unions of Quotients of Smooth Equivalence Relations". With Stephen Jackson. *Annals of Pure and Applied Logic*, 172 (2021), no 8.
(23) "$L(\mathbb{R})$ with Determinacy Satisfies the Suslin Hypothesis". With Stephen Jackson. *Advances in Mathematics*, 346 (2019), 305-328.
(24) "Ordinal Definability and Combinatorics of Equivalence Relations". *Journal of Mathematical Logic*, 19 (2019), no 2.
(25) "Definable Combinatorics of Some Borel Equivalence Relations". With Connor Meehan. Submitted.
(26) "When an Equivalence Relation with All Borel Classes will be Borel Somewhere?". With Menachem Magidor. *Proceedings of the American Mathematical Society*, 150 2 (2022), 833-847.
(27) "The Countable Admissible Ordinal Equivalence Relation". *Annals of Pure and Applied Logic*, 168 (2017), 1224-1246.
(28) "Equivalence Relations Which Are Borel Somewhere". *Journal of Symbolic Logic*, 82 (2017), no. 3, 893-930.

5. Talks

- Simon Semester, Gödel Program. IMPAM. Warsaw, Poland. 2026.
- European Set Theory Conference. University of Helsinki. Helsinki, Finland. 2026.
- Set Theory in the United Kingdom 19, University of Leeds. Leeds, United Kingdom. 2026.
- Cross-Alps Logic Seminar. 2026.
- 18th Luminy Workshop on Set Theory. Centre International de Rencontres Mathématiques. Marseille, France. 2025.
- Berkeley Conference on Inner Model Theory. University of California Berkeley. Berkeley, California. 2025.
- Set Theory Seminar. University of Vienna. Vienna, Austria. 2025.
- Cornell Logic Seminar. Cornell University. Ithaca, New York. 2025.
- Caltech Logic Seminar. California Institute of Technology. Pasadena, California. 2025.
- Rutgers Logic Seminar. Rutgers University. New Brunswick, New Jersey. 2025
- Set Theory Workshop. Mathematisches Forschungsinstitut Oberwolfach. Oberwolfach, Germany. 2025.
- Logic Colloquium. Kurt Gödel Research Center, University of Vienna. Vienna, Austria. 2024.

- Generalized Computability Theory. Centro Internacional de Encuentros Mathemáticos. Castro Urdiales, Cantabria, Spain. 2024.
- Determinacy, Inner Models, and Forcing Axioms. Erwin Schrödinger Institute. Vienna, Austria. 2024.
- Irvine Conference on Inner Model Theory. University of California at Irvine. Irvine, California. 2023.
- Mini Workshop in Set Theory. TU Wien, Vienna, Austria. 2023.
- Young Set Theory Workshop. University of Münster. Münster, Germany. 2023.
- 2023 BLAST Conference. University of North Carolina, Charlotte. Charlotte, North Carolina. 2023.
- Caltech Logic Seminar. California Institute of Technology. Pasadena, California. 2023.
- Set Theory Special Session. ASL North America Meeting. Irvine, California. 2023
- Nankai Logic Colloquium. Nankai University. Tianjin, China. 2022.
- Set Theory Workshop. Erwin Schrödinger International Institute. Vienna, Austria. 2022.
- Münster Conference on Inner Model Theory. University of Münster. Münster, Germany. 2022.
- Barcelona Set Theory Seminar. University of Barcelona. Barcelona, Spain. 2022.
- Set Theory Seminar. CUNY Graduate Center. New York City, New York. 2022.
- Set Theory Workshop. Mathematisches Forschungsinstitut Oberwolfach. Oberwolfach, Germany. 2022.
- Logic and Its Applications. Canadian Mathematical Society Winter Meeting. Montreal, Quebec, Canada. 2020.
- Caltech Logic Seminar. California Institute of Technology. Pasadena, California. 2020.
- University of Chicago Logic Seminar. University of Chicago. Chicago, Illinois. 2020.
- University of Wisconsin-Madison Colloquium. University of Wisconsin-Madison. Madison, Wisconsin. 2020.
- University of Wisconsin-Madison Logic Seminar. University of Wisconsin-Madison. Madison, Wisconsin. 2020.
- Choiceless Set Theory and Related Areas. Joint Mathematics Meeting. Denver, Colorado. 2020.
- UIUC Logic Seminar. University of Illinois Urbana-Champaign. Champaign, Illinois. 2019.
- Caltech-UCLA Logic Seminar. University of California Los Angeles. Los Angels, California. 2019.
- Berkeley Conference on Inner Model Theory. University of Califoria at Berkeley. Berkeley, California. 2019.
- Boise Extravaganza in Set Theory. Annual Meeting of AAAS Pacific Division. Southern Oregon University. Ashland, Oregon. 2019.
- The Core Model Induction and Other Inner Model Theoretic Tools. Rutgers University, New Brunswick, New Jersey. 2019.
- Mid-Atlantic Mathematical Logic Seminar (MAMLS 2018). Rutgers University. New Brunswick, New Jersey, 2018.
- First Girona Conference on Inner Model Theory. University of Girona. Girona, Catalonia. 2018.
- Set Theory Research Seminar. University of Münster. Münster, Germany. 2018.
- Set Theory Special Session. ASL North America Meeting. University of Western Illinois. Macomb, Illinois. 2018.
- South Eastern Logic Symposium. University of Florida. Gainesville, Florida. 2018.
- Rutgers Logic Seminar. Rutgers University. New Brunswick, New Jersey. 2018.
- South Eastern Logic Symposium. University of Florida. Gainesville, Florida. 2017.
- Set Theory Workshop. Mathematisches Forschungsinstitut Oberwolfach. Oberwolfach, Germany. 2017.
- Descriptive Inner Model Theory and HOD Mice. University of California at Irvine. Irvine, California. 2016.
- Boise Extravaganza in Set Theory. Annual Meeting of AAAS Pacific Division. University of San Diego. San Diego, California. 2016.
- Descriptive Set Theory and its Applications. AMS Spring Western Sectional Meeting. University of Utah. Salt Lake City, Utah. 2016.
- South Eastern Logic Symposium. University of Florida. Gainesville, Florida. 2016.

- Logic in Southern California. University of California at Irvine. Irvine, California. 2015.

6. Research Visits and Workshops

- Research on Better Understanding Solovay-Type Models. University of Leeds. Leeds, United Kingdom. 2026.
- Nairian Models Workshop. IMPAM. Warsaw, Poland. 2026.
- Reverse Mathematics. Erwin Schrödinger Institute. Vienna, Austria. 2025.
- Set Theory Workshop. Mathematisches Forschungsinstitut Oberwolfach. Oberwolfach, Germany. 2025.
- Vienna Inner Model Theory 2024. Technical University of Vienna. Vienna, Austria. 2024
- Determinacy, Inner Models, and Forcing Axioms. Erwin Schrödinger Institute. Vienna, Austria. 2024.
- Irvine Conference on Inner Model Theory. University of California at Irvine. Irvine, California. 2023.
- Young Set Theory Workshop. University of Münster. Münster, Germany. 2023.
- Invariant Descriptive Computability Theory Workshop. American Institute of Mathematics. San Jose, California. 2022.
- Set Theory Workshop. Erwin Schrödinger International Institute. Vienna, Austria. 2022
- Münster Conference on Inner Model Theory. University of Münster. Münster, Germany. 2022.
- Set Theory Workshop. Mathematisches Forschungsinstitut Oberwolfach. Oberwolfach, Germany. 2022.
- Münster Conference on Inner Model Theory. University of Münster. Münster, Germany. 2021. (Cancelled)
- Münster Conference on Inner Model Theory. University of Münster. Münster, Germany. 2020. (Cancelled.)
- Set Theory Workshop. Mathematisches Forschungsinstitut Oberwolfach. Oberwolfach, Germany. 2020. (Cancelled.)
- Berkeley Conference on Inner Model Theory. University of Califoria at Berkeley. Berkeley, California. 2019.
- The Core Model Induction and Other Inner Model Theoretic Tools. Rutgers University, New Brunswick, New Jersey. 2019.
- First Girona Conference on Inner Model Theory. University of Girona. Girona, Catalonia. 2018.
- University of Münster. Münster, Germany. 2018.
- Semester on Descriptive Set Theory and Polish Groups. Descriptive Set Theory Conference. Bernoulli Center. Lausanne, Switzerland. 2018.
- Set Theory Workshop. Mathematisches Forschungsinstitut Oberwolfach. Oberwolfach, Germany. 2017.
- Descriptive Inner Model Theory and HOD Mice. University of California at Irvine. Irvine, California. 2016.

7. Supervision

- Corentin Lagadec, Ph.D, Mathematics, TU Wien, Current.
- Julie Thompson, Master's, Mathematics, University of North Texas, 2023.

8. Teaching

- 104.703 Infinite Games. Fall 2025. TU Wien.
- 104.690 Axiom of Determinacy. Spring 2025. TU Wien.
- 104.649 Determinacy. Fall 2023. TU Wien.
- MATH 3510 Abstract Algebra I. Spring 2023. University of North Texas.
- MATH 2730 Multivariable Calculus. Fall 2022. University of North Texas.
- 21-400 Intermediate Logic. Spring 2022. Carnegie Mellon University.
- 21-112 Calculus II. Fall 2021. Carnegie Mellon University.

- 21-259 Calculus in Three Dimensions. Spring 2021. Carnegie Mellon University.
- 21-373 Algebraic Structures. Fall 2020. Carnegie Mellon University.
- MATH 4500/5600 Introduction to Topology. Spring 2019. University of North Texas.
- MATH 6010. Topics in Logic and Foundations. Fall 2018. University of North Texas.

9. Awards

- Fang Family Postdoctoral Fellowship. Carnegie Mellon University. 2020.
- University of North Texas Postdoctoral Research Award. University of North Texas. 2018.
- Scott Russell Johnson Graduate Dissertation Prize. California Institute of Technology. 2017.
- Scott Russell Johnson Prize for Excellence in Graduate Research. California Institute of Technology. 2016.
- Scott Russell Johnson Prize for Excellence in First Year Graduate Studies. California Institute of Technology. 2013.
- NSF RTG Graduate Fellowship. California Institute of Technology. 2012-2014.

10. Service

Conference, meeting, and seminar oranization.

- Descriptive Set Theory, Borel Dynamics, and Borel Combinatorics. Banff International Research Station. Institute for Advanced Study in Mathematics. Hangzhou, China. 2026.